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1. let $L([x,y]^T)=\begin{bmatrix}a_1x+b_1y+c_1\\ a_2x+b_2y+c_2\end{bmatrix}$

$\because L([1,2]^T)=\begin{bmatrix}a_1+2b_1+c_1\\ a_2+2b_2+c_2\end{bmatrix}=\begin{bmatrix}-2\\3\end{bmatrix}$, $L([1,-1]^T)=\begin{bmatrix}a_1-b_1+c_1\\ a_2-b_2+c_2\end{bmatrix}=\begin{bmatrix}5\\2\end{bmatrix}$

$\therefore \begin{cases} b_1=-\frac{7}{3}\\ b_2=\frac{1}{3}\\ a_1+c_1=\frac{8}{3}\\ a_2+c_2=\frac{7}{3}\end{cases}$

$\because$ Is a linear operator

$\therefore L([2,13]^T)=L(2[1,\frac{13}{2}]^T)=2L([1,\frac{13}{2}]^T)=2\times\begin{bmatrix}a_1+\frac{13}{2}b_1+c_1\\ a_2+\frac{13}{2}b_2+c_2\end{bmatrix}$

$=2\times\begin{bmatrix}\frac{8}{3}+\frac{13}{2}\times(-\frac{7}{3})\\ \frac{7}{3}+\frac{13}{2}\times\frac{1}{3}\end{bmatrix}=2\begin{bmatrix}-\frac{25}{2}\\ \frac{9}{2}\end{bmatrix}=\begin{bmatrix}-25\\9\end{bmatrix}$

2. (a) $L(\alpha X)=[\alpha x_1,\alpha x_2,2]^T\neq\alpha L(X)$ $\therefore$ Not linear transformation

(b) $L(\alpha X+\beta Y)=[\alpha x_1+\beta y_1,\ \alpha x_2+\beta y_2-\alpha x_1-\beta y_1,\ \alpha x_1+\beta y_1+2\alpha x_2+2\beta y_2]^T$

$=\alpha[x_1,x_2-x_1,x_1+2x_2]^T+\beta[y_1,y_2-y_1,y_1+2y_2]^T=\alpha L(X)+\beta L(Y)$

$\therefore L(X)$ is a linear transformation

3. (a) $L(P(x))=xP(x)$

$\because L(\alpha_1P(x_1)+\alpha_2P(x_2))=x_1\alpha_1P(x_1)+x_2\alpha_2P(x_2)=\alpha_1x_1P(x_1)+\alpha_2x_2P(x_2)$

$=\alpha_1L(P(x_1))+\alpha_2L(P(x_2))$ $\therefore$ Is linear

(b) $L(P(\alpha x))=\alpha^2x^2P(\alpha x)\neq\alpha L(P(x))$

$\therefore L(P(x))$ is not a linear transformation

4. $L(\alpha f+\beta g)=\int_0^x\alpha f(t)dt+\int_0^x\beta g(t)dt=\alpha\int_0^x f(t)dt+\beta\int_0^x g(t)dt=\alpha L(f)+\beta L(g)$

$\therefore L(f)$ is a linear transformation

$L(e^x)=\int_0^x e^t dt=e^x-1$

$L(x^2)=\int_0^x t^2dt=\frac{x^3}{3}$

5. (a) $I(e_3)=y_3$, $I(e_2)=y_2-y_3$ $I(e_3)=y_1-y_2$

(b) $A=\begin{bmatrix}0&0&1\\0&1&-1\\1&-1&0\end{bmatrix}$

6. (a) $\ker(L) = \{[0,0,0]^T\}$

(b) $\ker(L) = \{[0,0,x_3]^T \mid x_3 \in \mathbb{R}\}$

(c) $\ker(L) = \{[0,x_2,x_3]^T \mid x_2, x_3 \in \mathbb{R}\}$

7. (a) $S = V^{-1}U = -1\begin{bmatrix} 0 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = -\begin{bmatrix} -1 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}$

(b) $A = SBS^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -4 & -1 \end{bmatrix}$

(c) $L(V_1) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $a_{11}V_1 + a_{21}V_2 = 1\cdot\begin{bmatrix} 2 \\ 1 \end{bmatrix} + (-4)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} = L(V_1)$

$L(V_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $a_{12}V_1 + a_{22}V_2 = 0\cdot\begin{bmatrix} 2 \\ 1 \end{bmatrix} + (-1)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = L(V_2)$

8. $L(e_1) = b_1 + b_3 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $L(e_2) = b_2 + b_3 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$\left[\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 2 & 2 \end{array}\right] \to \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right] \quad \therefore A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

9. $L(e_1) = b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $L(e_2) = b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $L(e_3) = -b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\left[\begin{array}{cc|ccc} 1 & -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 \end{array}\right] \to \left[\begin{array}{cc|ccc} 1 & -1 & 1 & -1 & 1 \\ 0 & 2 & 0 & 2 & -2 \end{array}\right] \to \left[\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 \end{array}\right] \quad \therefore A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

10. $T(a_0 + a_1 t + a_2 t^2) = a_0 + a_1(3t-2) + a_2(3t-2)^2 = (a_0 - 2a_1 + 4a_2) + 3(a_1 - 4a_2)t + 9a_2 t^2$

$$= \begin{bmatrix} a_0 - 2a_1 + 4a_2 \\ 3a_1 - 12a_2 \\ 9a_2 \end{bmatrix} t = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & -12 \\ 0 & 0 & 9 \end{bmatrix}\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} t$$

$$\therefore A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & -12 \\ 0 & 0 & 9 \end{bmatrix}$$

11. $V = I^{-1}V = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix}$

$$B = V^{-1}AV = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix}\begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

12. $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

(a) $L(x) = Ax = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(b) $L(x) = Ax = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$

(c) $L(x) = Ax = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -15 \\ 9 \\ 6 \end{bmatrix}$

13. $T(-x_1, -x_2) = (-x_1 - 4|x_2|, -x_1 + 4x_2)$

$\neq (-x_1 + 4|x_2|, -x_1 + 4x_2) = -T(x_1, x_2)$

$\therefore$ No linear